

1 Approximate DP (ϵ, δ)

A mechanism M is (ϵ, δ) -**approximately DP** if for all neighbouring X, X' and all output sets S :

$$\Pr[M(X) \in S] \leq e^\epsilon \cdot \Pr[M(X') \in S] + \delta$$

δ = a small **leeway / slack** that allows the pure ϵ -DP guarantee to **fail with small probability** (Q1). Pure ϵ -DP is the special case $\delta=0$.

Gaussian mechanism (the natural mechanism for approx-DP):

- Add **Gaussian noise** $N(0, \dots)$ instead of Laplace
- Scales with the **L2-sensitivity**; for high-dim outputs it adds noise **proportional to $\sqrt{D}$** across D outputs (Q2) — vs Laplace which adds noise linearly proportional to D
- **Advantage: better utility due to adding quadratically less noise** in high dimensions (Q4)

Assignment: δ = leeway for small-probability events where ϵ -DP may fail (Q1); Gaussian noise scales **proportional to $\sqrt{D}$** (Q2); Gaussian advantage = **better utility / quadratically less noise** (Q4).

2 Properties of Approximate DP

Property	What it means
Post-processing	Any data-independent computation applied to a DP output stays DP (won't make it less private) — you can't undo privacy by processing the output
Basic composition	k mechanisms each (ϵ, δ) -DP $\rightarrow$ combined is $(k\epsilon, k\delta)$ -DP
Advanced composition	Combined privacy grows like $\epsilon\sqrt{(2k \ln(1/\delta))} + k\epsilon(e^\epsilon - 1)$ — much tighter than basic for large k

Assignment Q3 (select all): properties of Approx-DP = **post-processing** AND **composition** (NOT 'privacy even when $\delta > 1/n$ ', NOT 'output distributions completely identical').

3 The Exponential Mechanism

Used when the output is a **choice** (not a number) — e.g. pick a **price** in a digital-goods auction. Given a **reward/score function** $S(X, r)$ for each possible output r :

Select r with **probability proportional to $e^{\epsilon S(X, r) / (2\Delta)}$** (higher-scoring outcomes are exponentially more likely). This is Q6.

Sensitivity Δ = max over neighbouring X, X' of $|S(X, r) - S(X', r)|$.

Guarantees:

- **Privacy:** the Exponential Mechanism is ϵ -**DP** (Q7)
- **Utility:** it does **not** guarantee the maximum revenue, but guarantees the revenue is **not significantly lower than the optimal** (Q8):
expected score $\geq \text{OPT} - (2\Delta/\epsilon) \ln(|H|)$

Digital-goods example: buyers have valuations; choose a price to maximise revenue while protecting each buyer's valuation. Revenue function is #buyers whose valuation $\geq$ price.

4 Fairness in ML — real-world motivation

- **COMPAS recidivism:** White defendants **labelled lower-risk but did re-offend** (47.7%); African-American **labelled higher-risk but did NOT re-offend** (44.9%)
- **Amazon** scrapped an AI recruiting tool biased against women
- **Word embeddings:** 'nurse/secretary/teacher' → woman; 'officer/commander' → man
- Google 'ugliest language' result; discriminatory online ad delivery (Latanya Sweeney)

ML pipeline: Data → Algorithm → Model → World — bias can enter at any stage. Recap: PCA (find principal components) & Logistic Regression ($P(y=1|x)=g(w \cdot x)$, sigmoid; no closed form → gradient descent). **Assignment Q10 (select all):** from the COMPAS data — **White defendants labelled lower-risk yet re-offended AND African-American defendants labelled higher-risk but did NOT re-offend**. Q11: profession linked to sexist prejudice toward men = Cameraman.

5 Fair Unsupervised Learning — Fair PCA

Ordinary PCA can have **unequal reconstruction error** across groups (e.g. male vs female faces). **Loss** of a group A under projection $U_A = \text{Error}_A(U_A) - \text{Error}_A(\hat{A})$.

Fair PCA problem: $\min_U \max((1/|A|)\text{Loss}_A, (1/|B|)\text{Loss}_B)$ — minimise the **worst** group's average loss.

Goal (Q12): find a projection where the **reconstruction losses of groups A and B are EQUAL** (the protected group is not reconstructed worse). A poly-time algorithm gets equal losses using at most **d+1** dimensions.

6 Fair Supervised Learning — fairness definitions

There is **no single definition** of fairness (statistical parity, group fairness, demographic parity, equal opportunity, equalized odds, predictive parity, calibration...). Key ones:

Definition	Meaning (ideal value)
Statistical / Demographic Parity	$\text{Parity}(M,C) = \Pr[M(x)=1 \mid x \in C] - \Pr[M(x)=1] $ — selection rate independent of the protected group. Ideal parity = 0 (Q9)
Equality of Opportunity	Equal true-positive rate across groups: $\Pr[M=1 y=1, C] = \Pr[M=1 y=1]$
Predictive Parity (Outcome Test)	Equal precision: $\Pr[y=1 M=1, C] = \Pr[y=1 M=1]$
Individual Fairness	Similar individuals get similar outcomes: $D(M(x), M(x')) \leq d(x, x')$

Assignment Q9: ideal Statistical-Parity value = 0.

7 Impossibility, Fair LogReg & Fair MAB

Impossibility result (Kleinberg et al.): you generally **cannot satisfy all fairness criteria at once** (equal opportunity, calibration, etc.) — inherent trade-offs in fair risk scores.

Fair Logistic Regression: minimise the logistic loss **subject to a parity constraint** $|g_D(w)| \leq \tau$ — g_D measures how correlated the predictions are with the protected attribute. Optimising over a **fairness constraint space** inside the loss landscape.

Fair Multi-Armed Bandits (e.g. Swiggy/Zomato assigning partners to orders):

- Goal: maximise expected reward over time
- **Fairness notion:** if partner-a is **truly better** than partner-b, then $P(\text{algorithm picks a}) \geq P(\text{picks b})$
- **Idea:** sample uniformly among all arms whose **confidence interval overlaps** with the current winner — don't starve arms you're unsure about